

Computational Sciences

Basic Git Commands III

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1 Merge request

Your team is developing a Python software project to solve a problem. Assume the following roles:

- **Contributor A:** Writes new functionality (e.g., a Python function).
- **Contributor B:** Reviews and merges the code.

Follow this procedure to make a merge request.

1. Create a new repository, then each member of the team clones the repository.

```
mkdir local_repository
```

```
cd local_repository
```

```
git init
```

```
git remote add origin https://git.imp.fu-berlin.de/username/NameRepository.git
```

```
git branch -M main
```

```
git pull origin main
```

2. Contributor A creates a new branch to work on a feature:

```
git branch contributorA
```

3. Contributor A creates a new file `sum.py`, for example:

```
# sum.py
def sum(a, b):
    return a + b
```

Then:

```
git add sum.py
```

```
git commit -m "Add a function to sum two numbers"
```

```
git push origin contributorA
```

4. Contributor A navigates on GitLab:
 - (a) Go to **Merge Requests** → **New Merge Request**.
 - (b) Select the **source branch** (e.g., `contributorA`) and the **target branch** (e.g., `main`).
 - (c) Click **Compare branches and continue**.
 - (d) Fill in the form.
5. Contributor B reviews the merge request by checking the code and leaving comments. He can close or approve the merge request.
6. After the merge the team members should pull the latest changes:

```
git checkout main
git pull origin main
```

2 Fork

In GitLab, a fork is a personal copy of a repository that exists under your own namespace. It allows you to experiment with changes without affecting the original (upstream) repository. Forking is commonly used in collaborative projects, especially in open-source development, where contributors do not have direct write access to the main repository.

A fork is independent, meaning it won't automatically update with changes from the upstream repository. You can sync your fork with the upstream repository by fetching and merging upstream changes into your fork.

1. Fork the repository on GitLab.

2. Clone the forked repository.

```
mkdir local_repository
```

```
cd local_repository
```

```
git init
```

```
git remote add origin https://git.imp.fu-berlin.de/username/NameRepository.git
```

```
git branch -M main
```

```
git pull origin main
```

3. Add the original repository as an upstream remote:

```
git remote add upstream <original-repo-url>
```

4. Fetch updates from upstream:

```
git fetch upstream
```

```
git merge upstream/main
```

5. Push your changes to your fork:

```
git push origin
```

6. Open a merge request in GitLab to propose your changes.